Test 1 today

Test 1 (during the tutorial) covers material from Notes 1-4 (chapter 1 of the textbook). It contains four multiple choice questions (similar to online quiz questions) and two questions which require detailed written steps (similar to tutorial questions).

Polynomials and Rational Functions

Example Let $f(x) = 2x - 1$, $g(x) = x^2 - x + 2$. Then

$$\begin{aligned} f(x)g(x) &= (2x-1)(x^2-x+2) \\ &= 2x^3 - 2x^2 + 4x - x^2 + x - 2 \\ &= 2x^3 - 3x^2 + 5x - 2. \end{aligned}$$

This is a polynomial of degree 3, which is also called a *cubic* polynomial.

A *polynomial* is a function of the form $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, where $a_n, a_{n-1}, \ldots, a_0$ are constants and $a_n \neq 0$, and the largest exponent n is called the *degree* of the polynomial.

If $f(x)$ is a polynomial of degree m and $g(x)$ is a polynomial of degree n, then $f(x)g(x)$ is a polynomial of degree $m + n$.

The domain of a polynomial is $(-\infty, \infty)$.

Polynomials and Rational Functions

The quotient of two polynomials is usually not a polynomial. For example, $\frac{1}{x}$ and $\frac{x}{x^2+1}$ are not polynomials.
A *rational function* is the quotient of two polynomials. The domain of a rational function $f(x)/g(x)$ is the set of real numbers where $g(x) \neq 0$.
Recall that $\mathbb{R}$ denotes the set of all real numbers, that is, $\mathbb{R} = (-\infty, \infty)$.
Example The function $f(x) = \frac{x}{x-2}$ has domain

$$Dom\{f\} = \{x \in \mathbb{R} : x \neq 2\} = (-\infty, 2) \cup (2, \infty).$$

The function $g(x) = \frac{1}{x^2-4}$ has domain

$$\begin{aligned} Dom\{g\} &= \{x \in \mathbb{R} : x \neq 2, -2\} \\ &= (-\infty, -2) \cup (-2, 2) \cup (2, \infty). \end{aligned}$$

The function $h(x) = \frac{x}{x^2+1}$ has domain $Dom\{h\} = \mathbb{R}$.

2.2 Examples

Example Suppose that a company manufactures x units of a product per month. If we denote the fixed cost as F, the variable cost as $V(x)$, and the total cost of operation per month as $C(x)$, then

$$C(x) = V(x) + F.$$

The profit function $P(x)$ is the difference between the total revenue $R(x)$ and the cost $C(x)$:
$P(x) = R(x) - C(x)$.

2.2 Examples

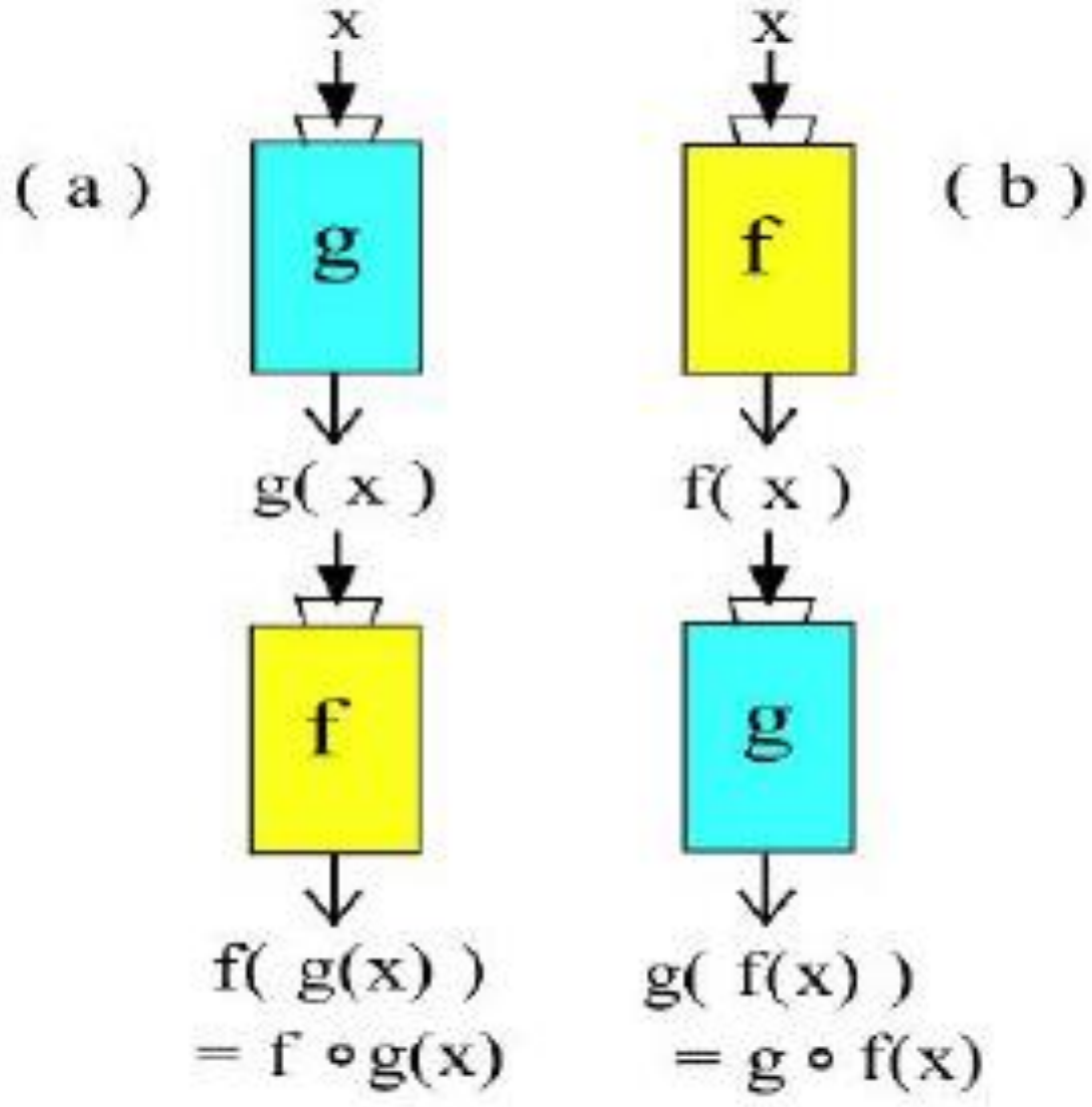
Example (Average Cost Function) A company makes chairs at the cost of $C(x) = 20x + 10000$ (dollars per month); where x is the number of chairs. The average cost $AC(x)$ is given by

$$AC(x) = \frac{C(x)}{x} = \frac{20x + 10000}{x}.$$

2.2 Compositions of Functions

Two functions f and g can be composed to form a new function $f \circ g$ such that $(f \circ g)(x) = f(g(x))$. The domain of $f \circ g$ consists of those numbers x in the domain of g for which $g(x)$ is in the domain of f, that is, such that $f(g(x))$ is well defined.
There are two ways to compose f and g: $f \circ g$ and $g \circ f$. In general $f \circ g$ and $g \circ f$ are not the same.

2.2 Compositions of Functions



2.2 Compositions of Functions

Example Suppose $f(x) = x^2$ and $g(x) = 2x + 1$. Then

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f(2x+1) = (2x+1)^2 \\ &= 4x^2 + 4x + 1, \\ (g \circ f)(x) &= g(f(x)) = g(x^2) = 2x^2 + 1.\end{aligned}$$

Example Suppose $f(x) = x^2$ and $g(x) = x^3$. Then

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f(x^3) = \left(x^3\right)^2 = x^6, \\ (g \circ f)(x) &= g(f(x)) = g(x^2) = \left(x^2\right)^3 = x^6.\end{aligned}$$

When we compute derivatives later, it is very important to recognize a complicated function as composition of simpler functions.

2.2 Compositions of Functions

Example Express the following functions as composition of simpler functions.
(i) $h(x) = \frac{1}{x^2+1}$, (ii) $h(x) = (x-4)^2$.
Solution (i) Let $f(x) = \frac{1}{x}$ and $g(x) = x^2 + 1$. Then

$$\begin{aligned} h(x) &= (f \circ g)(x) = f(g(x)) \\ &= \frac{1}{g(x)} = \frac{1}{x^2+1}. \end{aligned}$$

(ii) Let $f(x) = x^2$ and $g(x) = x - 4$. Then

$$\begin{aligned} h(x) &= (f \circ g)(x) = f(g(x)) \\ &= (g(x))^2 = (x-4)^2. \end{aligned}$$

2.2 Compositions of Functions

Example Suppose that a factory's cost of manufacturing x machine parts is $C(x) = 2200 + 48x$ dollars. An assembly line produces $N(t) = 8t - \frac{1}{2}t^2$ parts in t hours, where $1 \leq t \leq 8$. Express the cost as a function of hours of operation t.
Solution We have $x = N(t)$ and

$$\begin{aligned} C(N(t)) &= 2200 + 48N(t) \\ &= 2200 + 48\left(8t - \frac{1}{2}t^2\right) \\ &= 2200 + 384 - 24t^2. \end{aligned}$$

The domain of the above function is $1 \leq t \leq 8$.

2.3 Supply and Demand Functions

Demand Function: In a free market economy, consumer demand is the quantity that consumers are willing and able to buy per period of time at various prices. Consumer demand of a particular product depends upon its unit price. If x denotes the quantity demanded and p the unit price, then $p = f(x)$ is called the *demand function*.

2.3 Supply and Demand Functions

The demand function $f(x)$ is a decreasing function: Larger demand implies lower price. The graph of a typical demand function is shown below.

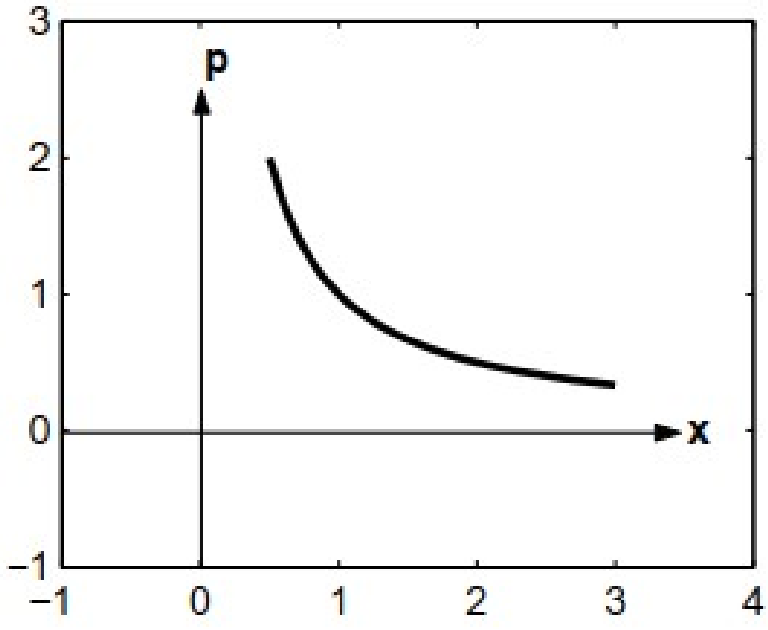


Figure 2.6: A demand curve $p = f(x)$.

2.3 Supply and Demand Functions

Supply Function: There also exists a relation between the price and the product's availability in the market. In general, an increase in the product's price stimulates the producer to increase the supply of the product, that is, the supply is an increasing function of the unit price.

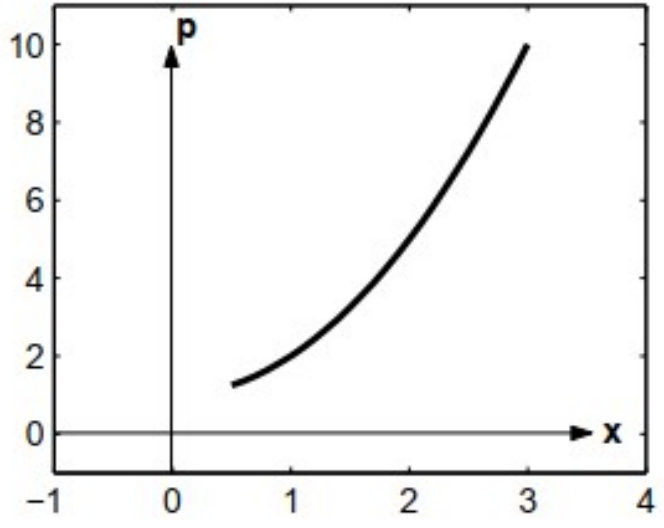


Figure 2.7: A supply curve $p = g(x)$.

2.3 Supply-Demand Equilibrium

Market Equilibrium The situation when the supply is equal to the demand is called market equilibrium. At market equilibrium we have

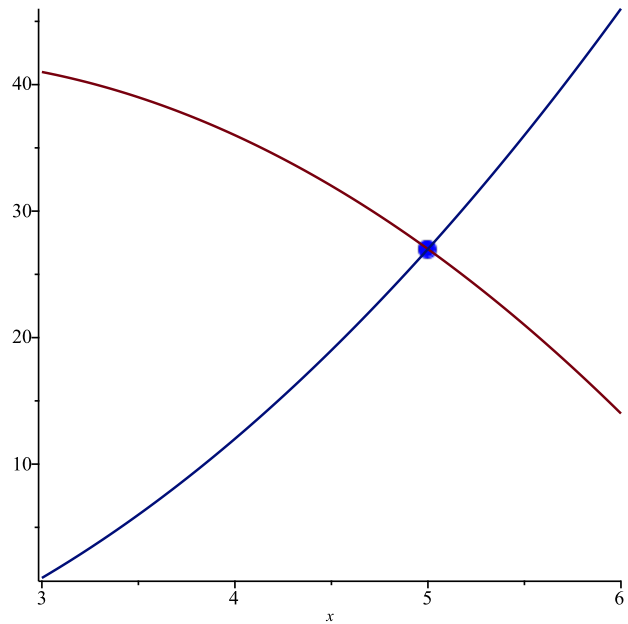
$$\begin{cases} p &= f(x), \\ p &= g(x) \end{cases}$$

the quantity x and price p satisfying the above system are called the *equilibrium quantity* and *equilibrium price*, respectively.

Example The demand function for a certain product is given by $f(x) = -2x^2 + 9x + 32$, and the supply function is given by $g(x) = 2x^2 - 3x - 8$, where p is in dollars and x is in units of a thousand. Find the equilibrium quantity and price.

2.3 Supply-Demand Equilibrium

Solution Solving the equation $-2x^2 + 9x + 32 = 2x^2 - 3x - 8$, i.e. $4x^2 - 12x - 40 = 0$, or $(x-5)(x+2) = 0$, we obtain $x = 5, -2$. Discard $x = -2$. Hence the equilibrium quantity is 5 and the equilibrium price is $g(5) = 27$.



2.3 Examples

Example The demand function for a new product is given by $p = f(x) = -3x + 20$; where p represents the selling price of the product, and x is the number sold in units of a thousand. The cost function is given by $C(x) = 5x + 12$ thousand dollars. What quantity of items sold will produce the maximum profit?
Solution The profit is given by the function

$$\begin{aligned} P(x) &= x(-3x + 20) - (5x + 12) \\ &= -3x^2 + 15x - 12 \\ &= -3\left(x - \frac{5}{2}\right)^2 + \frac{27}{4}. \end{aligned}$$

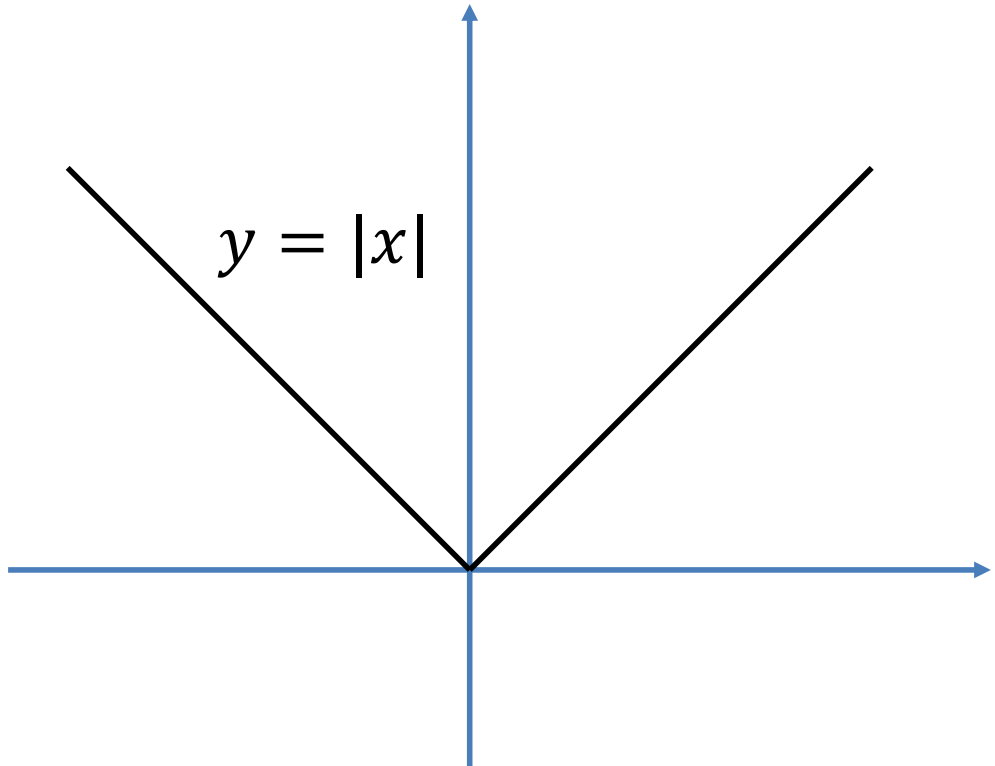
Hence the maximum profit is $27/4$ which occurs at $x = 5/2 = 2.5$. That is, 2500 new products should be sold to produce the maximum profit.

2.4 Piecewise-defined Functions

Recall the absolute value function

$$|x| = \begin{cases} -x & \text{if } x < 0, \\ x & \text{if } x \geq 0. \end{cases}$$

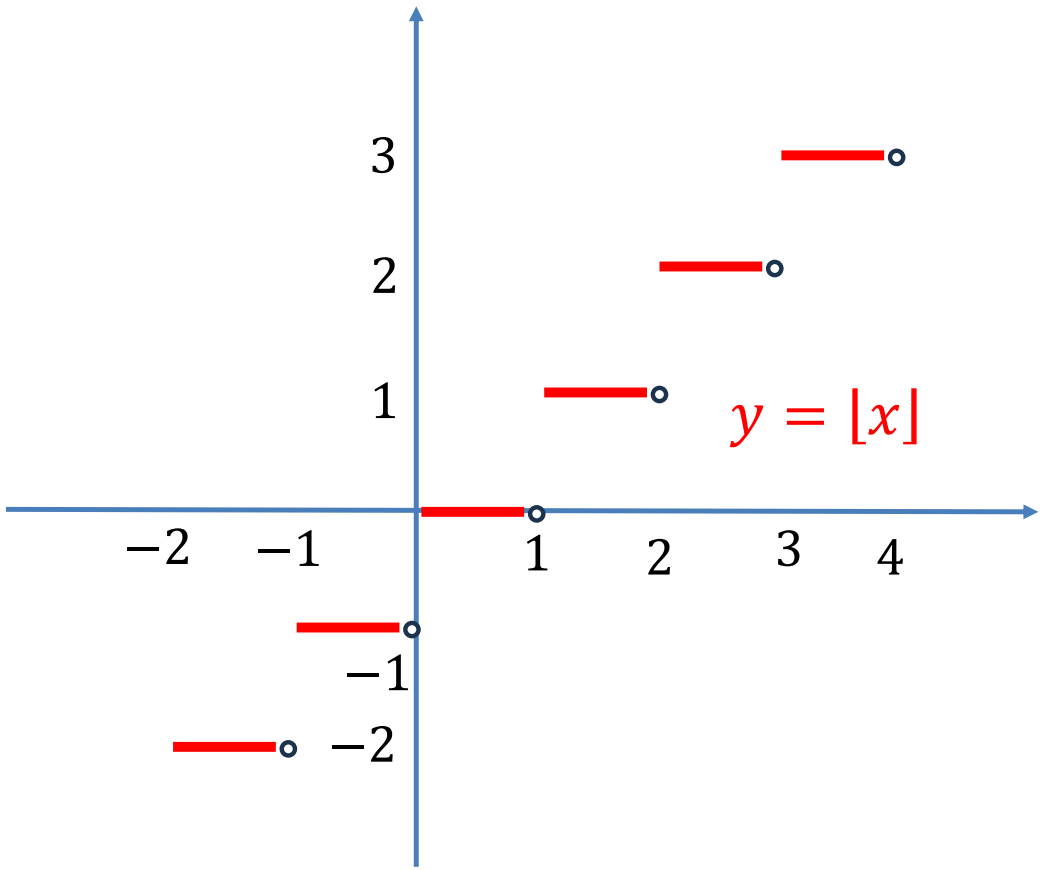
It is an example of *piecewise-defined functions*. Its graph is shown below.



2.4 Piecewise-defined Functions

Recall that $\lfloor x \rfloor$ denotes the nearest integer not exceeding x (round x down). This is called the *floor* function, which can be expressed as a piecewise-defined function (with many pieces):

$$\lfloor x \rfloor = n : n \leq x < n + 1,\ n \text{ is any integer.}$$



2.4 Piecewise-defined Functions

The ceiling function $\lceil x \rceil$ (round x up) can also be expressed as a piecewise-defined function:

$$\lceil x \rceil = n : n < x \leq n + 1,\ n \text{ is any integer.}$$

